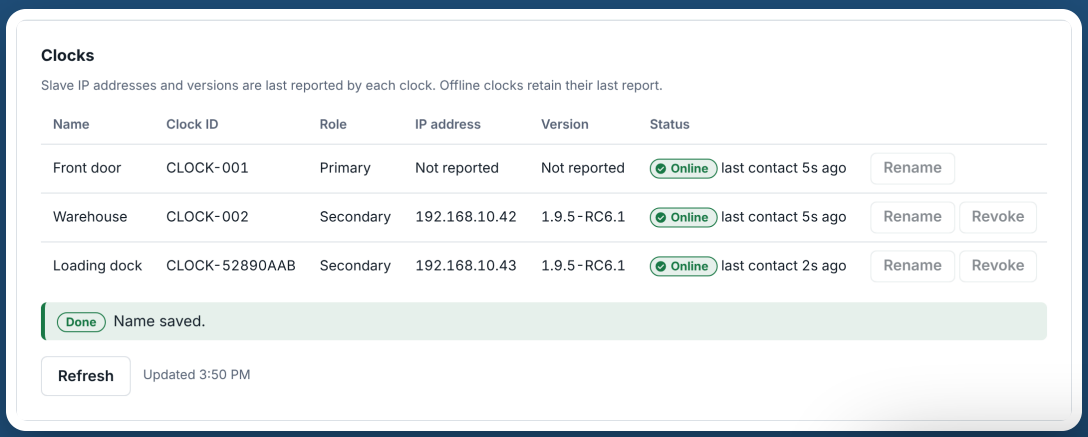
ICUBE DEVELOPMENT (CALGARY) LTD.

Timeclock Slave Clock Guide

Installing an extra clock, joining it to the Primary, and keeping it running: updates, outages, backups and replacements.





mk-timeclock Lite 1.9.5-RC6.1

For IT and installers

START HERE

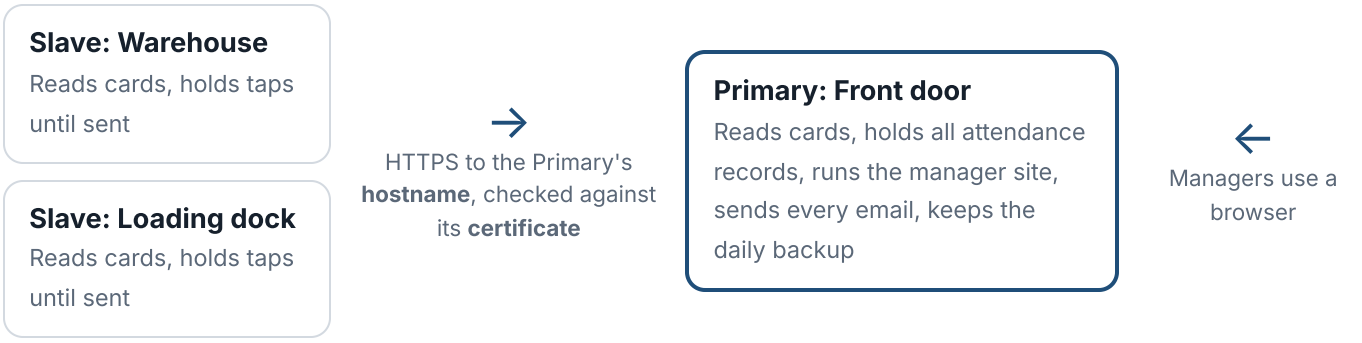
Why add a Slave clock

A Slave clock is an extra place for employees to tap in and out. It lets one timeclock system cover a building with several entrances, a second floor, a warehouse or a yard, with every tap ending up in the same records.

What it's for

- **Tap where you arrive.** Put a clock at each door or work area people actually use, instead of making everyone walk to one clock. Fewer forgotten taps and fewer corrections.
- **One set of records.** Every Slave sends its taps to the Primary. Attendance, timesheets, vacation, the payroll report and emails all come from the Primary, so it doesn't matter which clock someone used.
- **One card per person.** An employee's card works on every clock. Add, replace or deactivate a card once on the Primary and every Slave follows.
- **One manager site.** Managers never sign in to a Slave. They see and manage every clock from the Primary, including each Slave's status, IP address and software version.
- **Keeps working through short outages.** If the network drops, a Slave keeps recording taps for known cards with their real times and sends them when the connection returns.
- **Any location.** A Slave doesn't have to be in the same building. It can be at a branch office, a job site or another city, even on the other side of the world, as long as it can reach the Primary's hostname (Slaves at other locations).
- **Settings per location.** Each clock can have its own name, late grace, repeat window and screen brightness, so a warehouse clock can differ from the front door.

How the clocks fit together



A Slave doesn't need anyone to look after it day to day. Once it has joined, it is managed entirely from the Primary. You only touch the Slave itself to install it, update it, or recover it.

> **When you don't need a Slave**
> If everyone passes the same door, one Primary clock is enough. Add Slaves for extra entrances, separate work areas or other sites, not as a spare in case the Primary fails. A Slave can't replace the Primary.

Slaves at other locations

A Slave can be anywhere: a second building across town, a branch in another province, or an office on another continent. Its taps go into the same records on the Primary as if it were down the hall.

Distance doesn't matter to the Slave. All it needs is a working network path to the Primary's hostname on port 443, and a valid certificate on the Primary. The traffic is small: a short report every 30 seconds and each tap as it happens.

Connecting a remote site

- **Use a site-to-site VPN** (for example WireGuard or IPsec between the routers) so the remote Slave joins the Primary's network. The Primary stays on your internal network and is never exposed directly to the internet.
- **The hostname must resolve at the remote site.** The Slave uses the same address as every other clock, for example *https://timeclock.example.com*. Make sure DNS at the remote site, or over the VPN, returns the Primary's internal address.
- **The certificate is the same.** The Slave checks the Primary's certificate exactly as a local one does. If you use an internal certificate authority, install it on the remote Slave too.
- **Joining is done at the remote site.** Someone there, or you over the VPN, opens the Slave's setup page on port 8090 and enters the code. Generate the code only when they're ready, because it lasts 10 minutes.

When the link drops

Internet and VPN links drop more often than an office network. The Slave keeps recording taps for known cards with their real times, shows OFFLINE, and sends them when the link returns. Managers get a not-reporting email after 3 minutes and a reconnected email afterwards. A new employee's card can't be used at the remote site until the link is back.

Time zones

WHAT	WHICH TIME ZONE IT USES
The time on the Slave's screen	The Slave's own time zone, set in **CONFIG.TXT** (Step 2). Set it to the clock's location so staff see their local time.
Recorded taps	Stored as exact moments, so they're correct whatever the zones.
Schedules, late and early flags, pay periods, reports and emails	The Primary's time zone. Enter remote employees' schedules in the Primary's time, and expect their timesheets to show Primary times.

> **Check a remote clock's schedules carefully**
> An employee two hours east who starts at 8:00 their time has a scheduled start of 6:00 in the Primary's time. Set their schedule that way, or they'll be flagged late or early.

What a Slave clock is

A Slave is an extra card reader. It records taps and sends them to the Primary clock, which holds all the records and the manager site.

The manager site calls a Slave a **Secondary** clock. The installer, the setup page and the service names use **Slave**. They mean the same thing.

	PRIMARY	SLAVE
Reads cards	Yes	Yes
Attendance database	Yes, the only one	No. A small queue of taps not yet sent, plus a copy of known cards.
Manager site	Yes (HTTPS)	No. Only a setup page until it joins.
Sends email	Yes	No. The Primary emails managers about Slave outages.
Settings	Set on its manager site	Come from the Primary: name, grace, repeat window, brightness, font, time format.
Backups	Daily, its own	Daily, its own, kept on the Slave only.

> **A Slave is not a backup Primary**
> If the Primary fails, a Slave can't take over, and its queue is not a copy of the attendance records. Slaves keep recording taps for known cards while the Primary is away, and send them when it's back.

What you need

- An assembled clock (BPI-M2 Zero, reader, display, buzzer), built as in the **Timeclock Build and Install Guide**
- A microSD card of at least 16 GB with the **bpi-zero-wbuild 3.13** image
- The **mk-timeclock-lite-RC6.1.zip** package, the same version as the Primary
- The 2.4 GHz Wi-Fi name and password
- A computer on the same network with SSH and a web browser
- A manager sign-in on the Primary
- A test attendance card that is already set up for an employee

Order of operations

Do these in order. Each step is explained on the following pages. Allow about 45 minutes, most of it waiting for first boot and the install.

#	STEP	WHERE	DONE WHEN
1	Check the Primary is ready	MANAGER SITE	It runs 1.9.5-RC6.1, has a valid HTTPS certificate, and its hostname resolves from the Slave's network.
2	Write the SD card and set Wi-Fi	YOUR COMPUTER	CONFIG.TXT is saved and the card is in the clock.
3	First boot and sign in	SSH	You can sign in as root and the image marker is correct.
4	Install the Slave software	SSH	The installer says **Slave installed. Adopt at http://<clock-ip>:8090/**
5	Generate a code	MANAGER SITE	A 24-character code is shown. It lasts 10 minutes.
6	Send the join request	SETUP PAGE	The Slave says **Waiting for approval on the Primary clock.**
7	Approve the clock	MANAGER SITE	The page says **Approved CLOCK-...**
8	Check approval	SETUP PAGE	The Slave says **Adopted. This clock now records attendance.**
9	Name it and set its options	MANAGER SITE	It has a name, and is Online with its IP address and version.
10	Test at the clock, then close the case	AT THE CLOCK	A real tap shows IN on the Slave and appears on the Primary.

Hostname and certificate first

Slaves talk to the Primary only by its hostname (never its IP address), and only when the Primary has a valid certificate for that hostname. Sort both out before Step 2.

Keep the code step and the approval step close together

A code works once and expires after 10 minutes. Have the Slave setup page open before you select **Generate code**. If it runs out, generate a new one.

STEP 1

Check the Primary is ready

Important: Slaves connect to the Primary by hostname, not IP address
Every Slave reaches the Primary at *https://* plus the hostname on the Primary's certificate, for example *https://timeclock.example.com*. Never enter the Primary's IP address. The name must resolve to the Primary from the Slave's network (company DNS or a router DNS entry).

Important: a valid certificate must be on the Primary
Slaves will not communicate with a Primary that has no valid HTTPS certificate. The certificate must be current, trusted, and issued for the hostname the Slaves use. Without it a Slave can't join, and a joined Slave stops reporting if the certificate expires or is replaced with an untrusted one.

Same version

Under **Settings > System health** on the Primary, **Version** must say **1.9.5-RC6.1**. Update the Primary first if it doesn't. A Slave and its Primary should always run the same release.

A trusted HTTPS certificate

The Slave only talks to the Primary over HTTPS, and it checks the certificate. The temporary self-signed certificate a new Primary starts with will not work.

- Under **Settings > HTTPS certificate**, a real certificate must be installed, and the Overview must not say **Temporary self-signed certificate in use**.
- Note the certificate's hostname, for example **timeclock.example.com**. That is the address you'll type on the Slave, as *https://timeclock.example.com*.
- The Slave must be able to look up that name and reach it on port 443 from its Wi-Fi network.
- If the certificate comes from your own internal certificate authority, install that authority on the Slave first (Troubleshooting).

Network

CONNECTION	PORT	WHEN
Slave to Primary	443 (HTTPS)	Always, to the Primary's hostname. Over the office network, or over a site-to-site VPN for a remote site.
Your browser to Slave	8090 (HTTP)	Only during setup. The page turns itself off once the clock has joined. Use it on a trusted network only.
Your computer to Slave	22 (SSH)	Installing, updating, backups.
Slave to time servers	123 (NTP)	Always. A clock that isn't synchronized refuses taps.

STEPS 2 AND 3

Prepare the SD card and first boot

Step 2: write the card YOUR COMPUTER

1. Write **bpi-zero-wbuild-3.13-trixie-minimal-hw-bpi-m2-zero.img.gz** to the microSD card with balenaEtcher. This erases the card. Cancel any Windows prompt to format it afterwards.
2. Reinsert the card and open **CONFIG.TXT** on the **BPIWBUILD** volume. Set the six lines with your own values:

```
SSID=OfficeWiFi
PSK=wifi-password-here
COUNTRY=CA
HIDDEN=false
TIMEZONE=America/Edmonton
ROOT_PASSWORD=Choose-A-Strong-Password-26
```

3. Use a 2.4 GHz network. Set TIMEZONE to the clock's own location, usually the same as the Primary (Slaves at other locations). Keep the root password safe; the image removes it from the file after setup.
4. Eject the card, put it in the clock, check the wiring, then apply power.

Step 3: first boot and sign in SSH

1. Allow about ten minutes. The display stays blank until the timeclock software runs.
2. Find the clock in your router's device list and **reserve its address**. The Primary shows the address the Slave reports, so a fixed address makes it easier to find later.
3. Sign in from Terminal or PowerShell and check the image:

```
ssh root@192.168.10.43
cat /etc/bpi-zero-wbuild-release     # should say 3.13-trixie-minimal-hw
ls -l /dev/spidev0.0 /dev/i2c-0 /dev/gpiochip0
```

Replace 192.168.10.43 with your clock's address. This guide uses it for the example clock "Loading dock". The SSH root password is separate from any manager sign-in.

STEP 4

Install the Slave software

Copy **mk-timeclock-lite-RC6.1.zip** to **/root** on the clock with WinSCP or `scp`, then run these as root:

SSH

```
apt-get update
apt-get install -y unzip python3
cd /root
unzip mk-timeclock-lite-RC6.1.zip
cd mk-timeclock-lite-RC6.1
./install-slave.sh
```

The installer checks the platform, installs what's missing, builds the card reader program on the clock and sets up its services. It takes several minutes. It ends with:

```
Slave installed. Adopt at http://<clock-ip>:8090/ on trusted LAN.
Installed role: slave
```

What the installer sets up

- The card reader, display and buzzer program (`timeclock-agent`), left stopped until the clock joins
- The setup page on port 8090 (`mk-timeclock-setup`), started now
- Time sync and time zone helpers, and a daily local backup, enabled when the clock joins

It doesn't install the attendance database, manager site, mail sender or web server. Those are only on the Primary.

If the installer stops

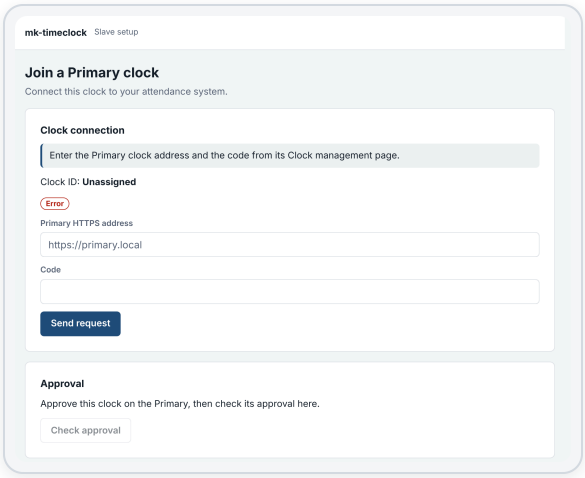
MESSAGE	MEANING
Primary database exists. Refusing Slave install to protect attendance records.	This card was used as a Primary. Use a fresh card, or move the Primary's data off first.
Existing role 'primary' differs from requested 'slave'.	Same cause. A clock can't change role in place.
Expected Debian armhf / BPI image marker missing / Missing /dev/...	Wrong image or interfaces not enabled. Rewrite the card with the 3.13 image.
Backup or restore is active.	Wait for it to finish and run the installer again.

STEPS 5 AND 6

Generate a code and send the request

Open the setup page first SETUP PAGE

On your computer, browse to **http://192.168.10.43:8090/** (your Slave's address). It shows **Join a Primary clock** with **Clock ID: Unassigned**.

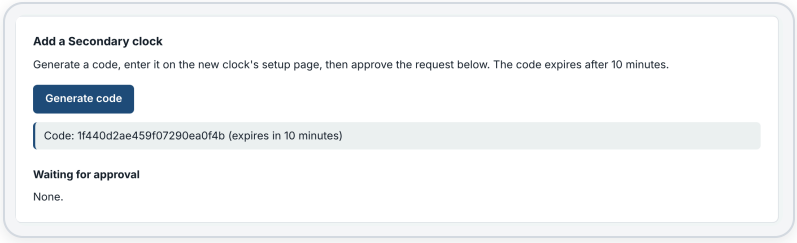


The Slave setup page before joining

> **The empty Error label**
> In RC6.1 the setup page always shows a red **Error** label under the Clock ID, even when nothing is wrong. Only read it as an error if text follows the label.

Step 5: generate a code MANAGER SITE

On the Primary, go to **Settings > Clock management** and select **Generate code** under **Add a Secondary clock**. The code is 24 characters (0 to 9 and a to f), works once, and expires after 10 minutes.



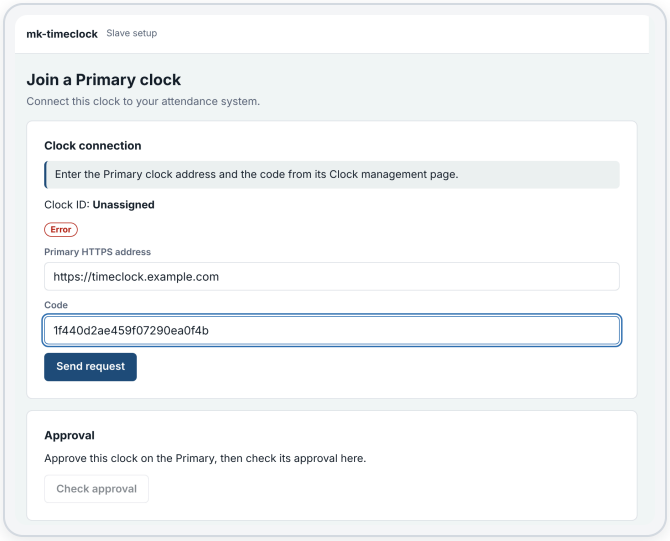
A new code on the Primary

Step 6: send the request SETUP PAGE

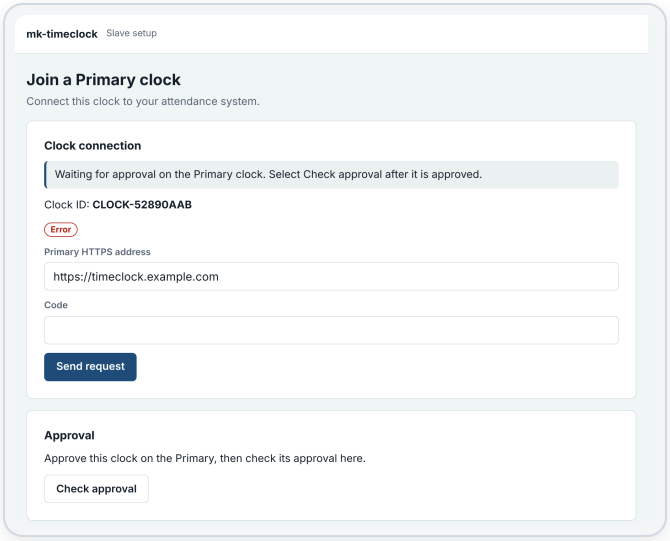
1. Enter the **Primary HTTPS address**: *https://* and the certificate's hostname, with no path, for example *https://timeclock.example.com*. **Use the hostname, not the IP address.**
2. Enter the **Code** and select **Send request**.
3. The Slave gives itself an ID such as **CLOCK-52890AAB** and says **Waiting for approval on the Primary clock.**

STEPS 6 AND 7

Request sent, then approve



Address and code entered



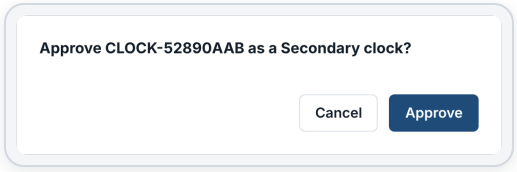
Waiting, with the Slave's new Clock ID

Step 7: approve the clock MANAGER SITE

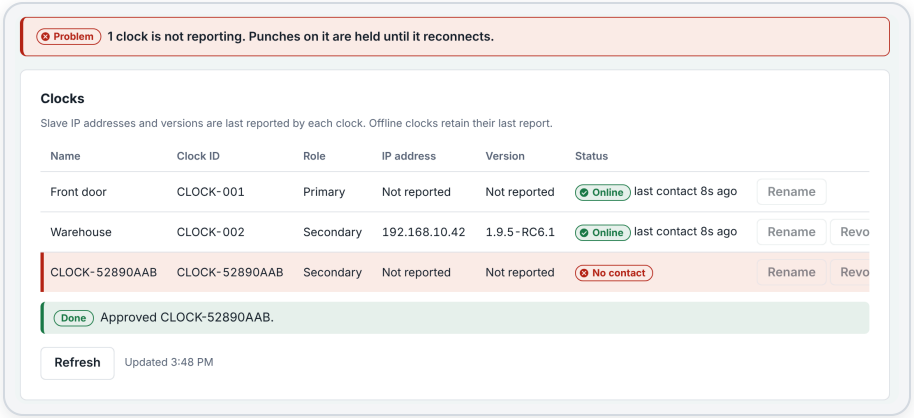
1. On **Clock management**, the request appears under **Waiting for approval** with the Slave's Clock ID. Select **Refresh** if it isn't there yet.
2. Check the Clock ID matches the one on the Slave's setup page, then select **Approve**.
3. Confirm **Approve CLOCK-52890AAB as a Secondary clock?**



The request waiting on the Primary



Confirming



Approved. It shows **No contact** until the Slave finishes setup

A new Slave gets three minutes to make first contact before it counts as not reporting. If you see **1 clock is not reporting**, finish Step 8.

STEP 8

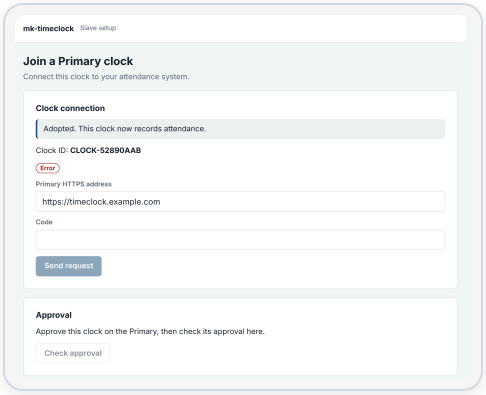
Check approval

1. Back on the Slave setup page, select **Check approval**.
2. The Slave collects its key from the Primary, proves it works with one authenticated contact, saves its settings, and starts the card reader and its daily backup.
3. The page says **Adopted. This clock now records attendance.**

A few seconds later the setup page switches itself off, so the address stops responding. That is normal. There's no settings page on a Slave after this; everything is managed from the Primary.

The clock's display shows the start-up logo, then the date, time and **TAP YOUR CARD**.

> **If Check approval fails part way**
>
> The Slave keeps the key it received, so you don't need a new code for a network or service problem. Reload the page; it says **Saved approval. Select Check approval to validate it and finish setup.** Select **Check approval** again. If the Primary rejects the key, the page says so and you start again from Step 5.



Joined



Starting up



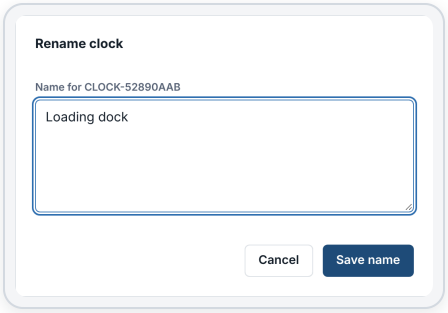
Ready for cards

STEPS 9 AND 10

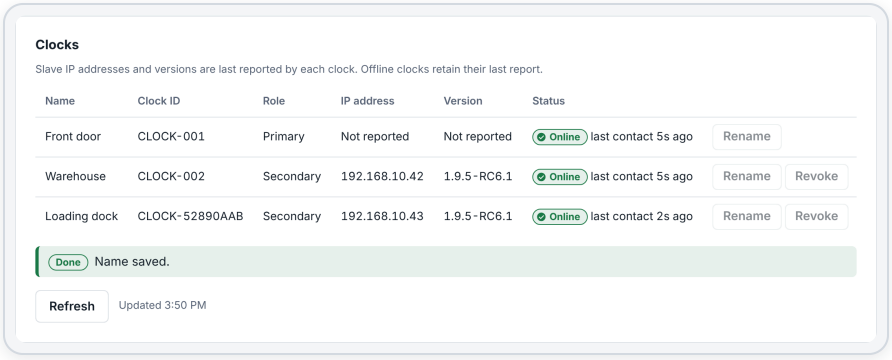
Name it, set it up and test it

Step 9: on the Primary MANAGER SITE

1. **Settings > Clock management**: select **Rename** on the new clock, type a name people will recognize, such as **Loading dock**, and select **Save name**. The name appears on every page and in outage emails.
2. Check the row: Online, its **IP address** and **Version 1.9.5-RC6.1**.
3. **Settings > Late grace and repeat taps**: the new clock starts with 5 minutes grace and a 120-second repeat window. Change them if this clock should match another.
4. **Settings > Clock display**: choose the new clock to set its brightness and dimming schedule. Font and 12 or 24-hour time follow the Primary automatically.



Naming the clock



Loading dock is online and reporting 192.168.10.43, running 1.9.5-RC6.1

Step 10: test at the clock AT THE CLOCK

- Tap an employee's test card. The clock shows **SAVED / CONFIRMING...**, then **IN** with the time and a beep. Tap again later for **OUT**.
- On the Primary, the tap appears on **Overview** and **Attendance**, recorded at this clock.
- Tap your manager card: **HELLO / MANAGER / NO PUNCH RECORDED**.
- Tap an unregistered card: **INVALID CARD / NOT RECORDED**, and it appears under **Settings > Unknown cards** with this clock's name.
- Undo the test taps on Attendance if they shouldn't count (Void, with a reason).

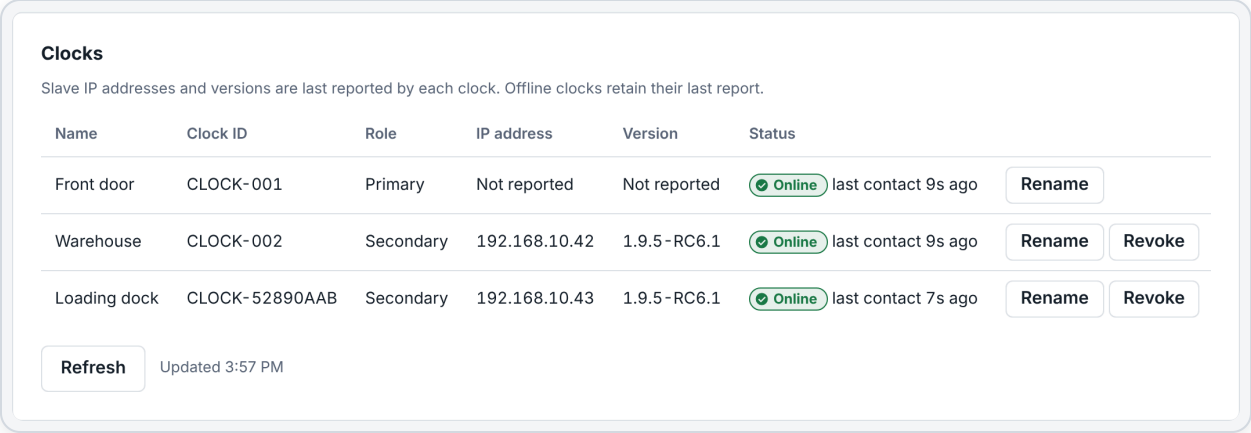




Then unplug power, fit the rear cover, mount the clock and power it back on. Check it returns to **TAP YOUR CARD** and is Online on the Primary.

Slaves on the Primary

Every joined Slave appears on the Primary under **Settings > Clock management**, with its **software version** and **IP address**.



Two Slaves, Warehouse and Loading dock, each showing its IP address and software version 1.9.5-RC6.1

COLUMN	WHAT IT SHOWS
Name	The name you gave the clock. Used on every page and in emails.
Clock ID	The ID the Slave chose when it joined, for example CLOCK-52890AAB. It never changes.
Role	**Primary** for the main clock, **Secondary** for each Slave.
IP address	The Slave's own address on your network, as it reports it every 30 seconds. Use it to find the clock for SSH or updates.
Version	The software the Slave is running, for example **1.9.5-RC6.1**. Check it after every update.
Status	Online with the time of the last report, or a warning such as Offline or 2 pending.

- The page refreshes every 15 seconds. Select **Refresh** to update it now.
- An offline Slave keeps showing the last IP address and version it reported.
- **Not reported** means the clock hasn't sent them. The Primary's own row always shows this, and so does a Slave still running software older than RC6.1.

> **The IP address is for finding the Slave, not for connecting to the Primary**
> The Slave reports its own address so you can locate it. The Slave itself always reaches the Primary by the Primary's hostname, over HTTPS with a valid certificate.

How a Slave behaves

Every 30 seconds the Slave sends the Primary a short report: that it's alive, how many taps it's holding, whether its time is synchronized, its IP address and its version. It also collects the card list and display settings.

When it loses the Primary

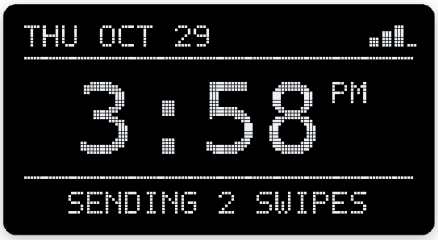
- Cards it already knows still record. Each tap is saved on the clock with its real time and shows **SAVED / CONFIRMING...** instead of IN or OUT.
- The idle screen says **OFFLINE** and counts the saved taps, for example **3 SWIPES SAVED**.
- After three minutes it chirps once, then every five minutes while disconnected. Set `BUZZER_OFFSET=none` to silence the buzzer completely.
- New cards, cards changed on the Primary since the outage started, and mode changes don't reach it until it reconnects.
- When the connection returns it shows **SENDING 2 SWIPES** until the queue is empty. Nobody needs to tap again.



Tap saved on the clock
Shown for every tap while offline



Offline
Three taps waiting



Reconnected
Sending the saved taps

When its time isn't synchronized

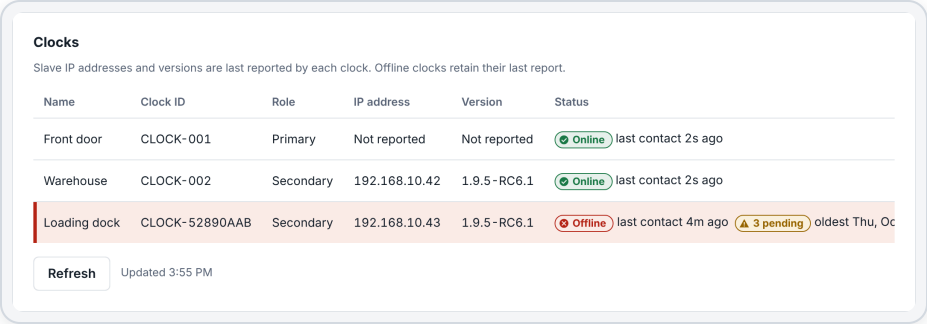
A clock that isn't sure of the time refuses every tap with **CLOCK NOT SYNCED / NOT RECORDED**. Check its network and its access to time servers. Changing the time zone doesn't fix this.

If the SD card fails before taps are sent, those taps are lost. Keep Slaves online, and check Clock management for a growing pending count.



When a Slave stops reporting

After three minutes without a report, the Primary marks the Slave (Offline), counts it under **Needs attention** on the Overview and in the Settings menu, and emails every active manager once. The pending count shown is the last one it reported.



Loading dock offline for 4 minutes, last reporting 3 taps waiting

EMAIL TO MANAGERS

Timeclock: Loading dock is not reporting

Clock Loading dock (CLOCK-52890AAB) has not reported for at least three minutes.
Last contact: 2026-10-29 15:51 UTC-0600
Last reported: 3 pending. Current count is unknown while disconnected.

Attendance may be incomplete. Check power, network and the clock before finalizing reports. Cached cards may still record locally, but this cannot be confirmed remotely.

EMAIL TO MANAGERS

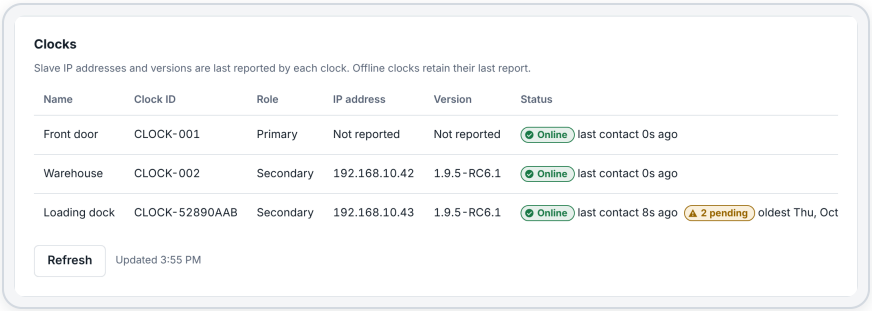
Timeclock: Loading dock reconnected

Clock Loading dock (CLOCK-52890AAB) is reporting again.
Last contact: 2026-10-29 15:55 UTC-0600
0 pending. Uploads may still be draining; check Clock management.

Check rejected uploads and out-of-order punches before treating attendance as complete.

After it reconnects

The row goes back to (Online) but keeps a (2 pending) warning until the queue is empty. Wait for it to clear before reviewing that pay period.



Back online, still sending 2 saved taps

What to check

Power and Wi-Fi	Is the display on? Does it say OFFLINE? Can you reach it with SSH?
The Primary's address	Has the Primary's hostname, address or certificate changed?
Rejected uploads	After a long outage, check **Settings > Rejected uploads** for refused or out-of-order taps, and add missed punches where needed.

MAINTENANCE

Updating a Slave

Update the Primary first, then every Slave, with the same release. A Slave on older software keeps working but doesn't report its address or version, and the Primary shows **Not reported**.

1. On the Primary's **Clock management**, check the Slave is Online with no pending taps. Pick a quiet time; the clock can't read cards while it updates. Other clocks keep working.
2. Copy the new zip to the Slave and run the installer from the new folder: SSH

```
cd /root
unzip mk-timeclock-lite-RC6.1.zip
cd mk-timeclock-lite-RC6.1
./install-slave.sh
```

3. An adopted Slave keeps its Clock ID, key, settings and any saved taps. The installer ends with **Adopted Slave installed; local RFID agent started.** There's no need to join it again.
4. On the Primary, check the row shows the new **Version** after the next report (about 30 seconds, then **Refresh**).
5. At the clock, tap a test card and check the display and buzzer. Do this after every update.

Useful checks on a Slave

```
systemctl status timeclock-agent.service              # the card reader program
journalctl -u timeclock-agent.service -n 50 --no-pager
timedatectl show -p NTPSynchronized                    # must say yes
/opt/icubedev-timeclock/bin/check-config.sh            # checks the saved settings
```

> **Don't edit the Slave's settings file by hand**
> /etc/timeclock/timeclock.env holds the Slave's ID, the Primary's address and its private key. A wrong value stops it reporting. To point a Slave at a different Primary, revoke it and join it again (Maintenance, continued).

Backups and replacing a Slave

Daily backups

Each joined Slave backs up its own queue, card list and settings every day at about 02:15, while it keeps reading cards. It keeps the last 30 in **/var/backups/timeclock** on the Slave only. These aren't on the Primary's Backup and restore page and aren't copied anywhere. Missed backups are skipped, not caught up.

```
/opt/icubedev-timeclock/bin/timeclock-backup backup     # back up now
/opt/icubedev-timeclock/bin/timeclock-backup list
/opt/icubedev-timeclock/bin/timeclock-backup verify /var/backups/timeclock/ARCHIVE.tar.gz
/opt/icubedev-timeclock/bin/timeclock-backup restore /var/backups/timeclock/ARCHIVE.tar.gz --yes
```

> **Slave backups contain its key**
> Each archive includes the Slave's private key. Treat a copy like a password. Restore only onto the same Slave. A restore refuses to run if the clock still has taps waiting, or if the archive belongs to another clock.

Replacing a Slave

The simple way (recommended): on the Primary, wait until the old clock has no pending taps, then **Revoke** it. Set up the new hardware from Step 2 and join it as a new clock with its own ID. Give it the old name.

Keeping the same Clock ID (only when the old clock had taps that never reached the Primary):

1. Keep the old clock powered off. Never run two clocks with the same identity.
2. Install the new clock to Step 4, then, as root, put back the old clock's saved configuration from its backup (the archive's **data/slave-recovery** folder), and reinstall any internal certificate authority.
3. Restore the old clock's backup with `timeclock-backup restore`, and confirm the Primary shows it Online and the pending count drains.

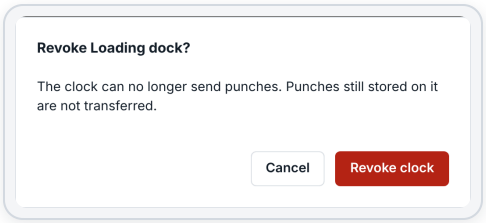
If the old clock's SD card failed, its local backups are likely gone too. Ask managers to add any missing punches by hand.

Revoking and re-joining

Revoking a Slave

On **Clock management**, select **Revoke** on the clock and confirm. The Primary stops accepting anything from it and the row shows ⊙ Revoked. Use this when a clock is retired, stolen, moved to another Primary, or being replaced.

> **Wait for pending taps first**
> Revoking does not transfer taps still stored on the clock. Check the row has no ⚠ pending count before you revoke.



Revoking Loading dock

Re-joining a revoked Slave

A revoked Slave keeps its old key, which no longer works. To join it again: SSH

```
systemctl disable --now timeclock-agent.service timeclock-backup.timer
rm /etc/timeclock/adopted
systemctl enable --now mk-timeclock-setup.service
```

1. Open **http://<clock-ip>:8090/**. It says **Saved approval. Select Check approval to validate it and finish setup.**
2. Select **Check approval**. The Primary rejects the old key and the page says **The Primary refused this clock key. Generate a new code on the Primary and Send request.** The old key is removed.
3. Continue from Step 5: generate a code, send the request, approve, check approval.

Deleting the marker alone doesn't let an old key back in. The Primary must approve a new request.

TROUBLESHOOTING

Setup page messages

THE SLAVE SETUP PAGE SAYS	WHAT TO DO
certificate verify failed	The Slave doesn't trust the Primary's certificate. Install a real certificate on the Primary, or install your internal certificate authority on the Slave (below).
Enter the Primary address as https://name, with no path.	Use *https://* plus the hostname only, for example https://timeclock.example.com.
The code is 24 characters (0-9, a-f).	Re-type or paste the code from Clock management.
The code expired. Generate a new one on the Primary.	More than 10 minutes passed, or the code was already used. Generate a new code and Send request again.
HTTP Error, Name or service not known, or timed out	The Slave can't reach the Primary. Check the hostname resolves on the Slave's network and port 443 is open.
The Primary approved a different clock. Request a new code.	The Clock IDs didn't match. Generate a new code and start again.
The Primary refused this clock key.	The clock was revoked or its key replaced. Generate a new code and Send request.
Saved approval. Select Check approval...	Setup was interrupted after approval. Select Check approval to finish.
This clock has already joined a Primary.	It's already set up. See Revoking and re-joining to move it.
This clock was not installed as a Slave.	Run `./install-slave.sh`.
A setup request is already running. Please wait.	Another request is in progress. Wait a few seconds and reload.
The page doesn't load at all	After joining this is normal. Before joining, check `systemctl status mk-timeclock-setup.service`.

Installing an internal certificate authority on the Slave

```
cp Company-Root-CA.crt /usr/local/share/ca-certificates/
update-ca-certificates
```

Use your authority's certificate in PEM format with a .crt name. Then select Send request again.

After joining

YOU SEE	WHAT TO DO
⊗ No contact on the Primary	The Slave was approved but hasn't reported. Finish Check approval, then check `timeclock-agent` is running.
⊗ Offline	No report for 3 minutes. Check power, Wi-Fi and the Primary's address (Running a Slave).
⊗ Clock not synchronized	The Slave's time isn't synchronized, so it refuses taps. Check its network access to time servers.
⊗ Local queue error	The Slave can't save taps on its SD card. Check free space and the card. Replace the card if needed.
⊗ 3 rejected locally	Taps the Primary refused, still held on the Slave. Look at **Settings > Rejected uploads** and add missed punches where needed.
⚠ 2 pending	Taps waiting to send. Normal for a moment after an outage. If it keeps growing, the Slave can't reach the Primary.
IP address and Version say Not reported	The Slave runs older software. Update it. The Primary's own row always says Not reported.
CARD READ, then INVALID CARD for a real employee	The card was added on the Primary while this Slave was offline. Wait for it to come back online, then tap again.
Display blank after install	Normal before joining. After joining, check the agent log and the display wiring.
No chirp, or a chirp you don't want	Check `BUZZER_OFFSET` and the buzzer wiring. `none` turns sound off.

Quick reference

Install or update	`./install-slave.sh`
Setup page (before joining)	http://<clock-ip>:8090/
Card reader service	`timeclock-agent.service`
Setup service	`mk-timeclock-setup.service`
Settings file	/etc/timeclock/timeclock.env (don't edit)
Joined marker	/etc/timeclock/adopted
Backups	/var/backups/timeclock, daily at about 02:15, last 30 kept
On the Primary	Settings > Clock management, Late grace and repeat taps, Clock display, Rejected uploads

> **About the examples**
> The screenshots come from the actual timeclock software (1.9.5-RC6.1) running on sample data. The names, addresses and card numbers are made up.